

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular and Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Electronics Engineering/ Electronics & Telecommunication Engineering / Electronics and Communication Engineering

Semester : VII

Subject Code & Name: BTEXOE704C/ BTETOE704C & Data Structure & Algorithms Using Java Programming

Max Marks: 60

Date:12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Objective type questions. (Compulsory Question)

12

- | | | | |
|---|--|----|---|
| 1 | What is a data structure? | L1 | 1 |
| | a. A programming language b. A collection of algorithms c. A way to store and organize data d. A type of computer hardware | | |
| 2 | Which of the following is an example of an Elementary Item? | L1 | 1 |
| | a. Student Name b. Student Roll Number c. Student Age d. Student Address | | |
| 3 | The time complexity of Linear Search in the worst case is: | L1 | 1 |
| | a. $O(n)$ b. $O(\log n)$ c. $O(n^2)$ d. $O(1)$ | | |
| 4 | In Binary Search, how do you determine the next search interval? | L1 | 1 |
| | a. Compare the middle element with the target and choose either the left or right half b. Compare all elements sequentially c. Divide the array into three equal parts d. Perform a linear scan to find the next interval | | |
| 5 | What is the primary principle followed by a Stack? | L2 | 1 |
| | a. FIFO (First In, First Out) b. LIFO (Last In, First Out) c. Random Order d. FILO (First In, Last Out) | | |
| 6 | Which operation checks the topmost element of a stack without removing it? | L3 | 1 |
| | a. Pop b. Push c. Peek d. Enqueue | | |
| 7 | Which of the following is NOT a type of linked list? | L3 | 1 |
| | a. Singly Linked List b. Doubly Linked List c. Circular Linked List d. Static Linked List | | |
| 8 | In a Doubly Linked List, how many pointers does each node have? | L2 | 1 |
| | a. One pointer to the next node b. Two pointers: one to the next node and one to the previous node c. Three pointers: next, previous, and parent d. No pointers | | |
| 9 | What is a leaf node in a tree? | L1 | 1 |
| | a. A node with no parent b. A node with exactly one child c. A node with no children d. A node with multiple parents | | |

10	In a binary tree, what is the maximum number of children a node can have?	L1	1
	a. 1 b. 2 c. 3 d. Unlimited		
11	What is the time complexity to access an element in an array using an index?	L3	1
	a. $O(n)$ b. $O(\log n)$ c. $O(1)$ d. $O(n^2)$		
12	If an array is declared as <code>int arr[10]</code> ; how many elements can it store?	L2	1
	a. 10 b. 11 c. 9 d. 8		
Q. 2	Solve the following.		12
A)	Define Searching, Explain Linear search method with the help of example; also give best, worst, average case of the same.	L1	6
B)	List different operations performed on Data structure. Explain any two with example.	L2	6
Q.3	Solve the following.		12
A)	Convert the infix expression $(A+B)*C/D$ to	L3	6
	1) Prefix expression		
	2) Postfix expression		
B)	Explain Stack data structure. Write an algorithm of PUSH and POP operations of stack.	L2	6
Q. 4	Solve Any Two of the following.		12
A)	Write an algorithm to insert node at the beginning in Single Linked list.	L3	6
B)	Write differences between array and linked list.	L1	6
C)	Explain representation of linked lists in the memory.	L2	6
Q.5	Solve Any Two of the following.		12
A)	What is a binary tree? Explain any two types of binary tree with an example.	L1	6
B)	Define the following terms with neat diagrams:	L2	6
	1) Siblings		
	2) Leaf Node		
	3) Ancestor of Node		
C)	Write Binary Tree Properties.	L2	6
Q. 6	Solve Any Two of the following.		12
A)	How does the Bubble sort works? Explain with the help of example.	L1	6
B)	Write an algorithm for Insertion sort.	L3	6
C)	What is a Graph? Explain the following terms	L2	6
	1) Undirected graph		
	2) Directed graph		
	3) Weighted and Unweighted graph		
	*** End ***		